

22MA201- ENGINEERING MATHEMATICS- II

UNIT III VECTOR DIFFERENTIAL CALCULUS

LP-13 TANGENT AND NORMAL VECTORS, PROJECTILE MOTION

Tangent Planes and Normal Lines

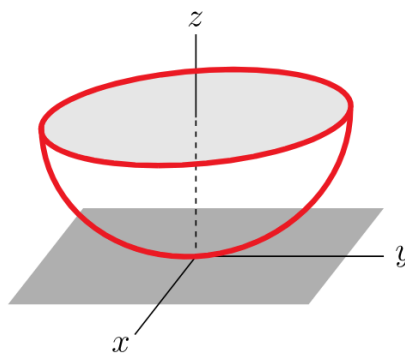
1.1.Tangent Planes

The tangent plane to a surface S at a point (x_0, y_0, z_0) is the plane that fits S best at (x_0, y_0, z_0) .

For example, the tangent plane to the hemisphere

$$S = \{(x, y, z) | x^2 + y^2 + (z - 1)^2 = 1, 0 \leq z \leq 1\}$$

at the origin is the xy -plane, $z = 0$.



Let $z = f(x, y)$ be a function of two variables. We can define a new function $F(x, y, z)$ of three variables by subtracting z . This has the condition

$$F(x, y, z) = 0.$$

Now consider any curve defined parametrically by

$$x = x(t), y = y(t), z = z(t).$$

We can write,

$$F(x(t), y(t), z(t)) = 0.$$

Differentiating both sides with respect to t , and using the chain rule gives

$$F_x(x, y, z)x' + F_y(x, y, z)y' + F_z(x, y, z)z' = 0$$

Notice that this is the dot product of the gradient function and the vector $\langle x', y', z' \rangle$,

$$\nabla F \cdot \langle x', y', z' \rangle = 0.$$

In particular the gradient vector is orthogonal to the tangent line of any curve on the surface. This leads to:

1.1.1. Definition: Tangent Plane

Let $F(x, y, z)$ define a surface that is differentiable at a point (x_0, y_0, z_0) , then the tangent plane to $F(x, y, z)$ at (x_0, y_0, z_0) is the plane with normal vector

$$\nabla F(x_0, y_0, z_0)$$

that passes through the point (x_0, y_0, z_0) . In particular, the equation of the tangent plane is

$$\nabla F(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0.$$

1.1.2. Example

Find the equation of the tangent plane to $z = 3x^2 - xy$ at the point $(1, 2, 1)$.

Solution

We let $F(x, y, z) = 3x^2 - xy - z$ then

$$\nabla F = \langle 6x - y, -x, -1 \rangle.$$

At the point $(1, 2, 1)$, the normal vector is

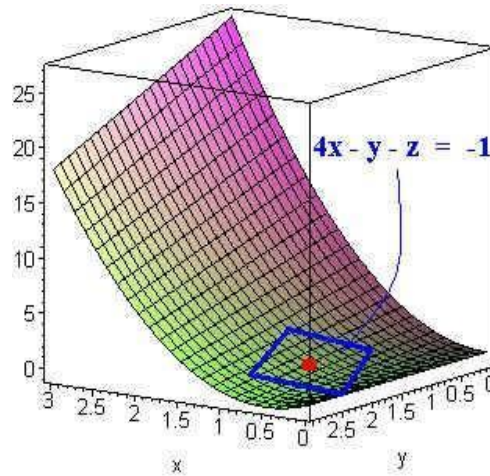
$$\nabla F(1, 2, 1) = \langle 4, -1, -1 \rangle.$$

Now use the point normal formula for a plane

$$\langle 4, -1, -1 \rangle \cdot \langle x - 1, y - 2, z - 1 \rangle = 0$$

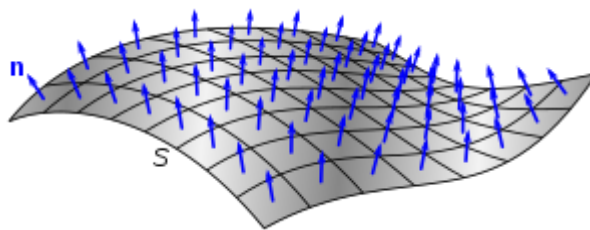
or $4(x - 1) - (y - 2) - (z - 1) = 0$.

Finally we get $4x - y - z = 1$.



1.2. Normal Lines

Given a vector and a point, there is a unique line parallel to that vector that passes through the point. In the context of surfaces, we have the gradient vector of the surface at a given point. This leads to the following definition.



1.2.1. Definition: Normal Lines

Let $F(x, y, z)$ define a surface that is differentiable at a point (x_0, y_0, z_0) , then the normal line to $F(x, y, z)$ at (x_0, y_0, z_0) is the line with normal vector

$$\nabla F(x_0, y_0, z_0).$$

that passes through the point (x_0, y_0, z_0) . In Particular the equation of the normal line is

$$x(t) = x_0 + F_x(x_0, y_0, z_0)t,$$

$$y(t) = y_0 + F_y(x_0, y_0, z_0)t,$$

$$z(t) = z_0 + F_z(x_0, y_0, z_0)t.$$

1.2.2. Example.

Find the parametric equations for the normal line to $x^2yz - y + z - 7 = 0$ at the point $(1,2,3)$.

Solution

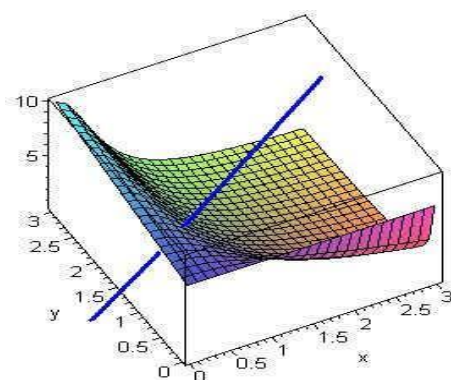
We compute the gradient:

$$\nabla F = \langle 2xyz, x^2z - 1, x^2y + 1 \rangle = \langle 12, 2, 3 \rangle.$$

Now use the formula to find

$$x(t) = 1 + 12t, \quad y(t) = 2 + 2t, \quad z(t) = 3 + 3t.$$

The diagram below displays the surface and the normal line.



Problems

1. Find the equation of the tangent plane and normal line to the surface $2x^2 + y^2 + 2z = 3$ at the point $(2, 1, -3)$.

Solution

We have, $f(x, y, z) = 2x^2 + y^2 + 2z = 3$

$$\frac{\partial f}{\partial x} = 4x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial z} = 2$$

At point $(2, 1, -3)$

$$\frac{\partial f}{\partial x} = 8, \quad \frac{\partial f}{\partial y} = 2, \quad \frac{\partial f}{\partial z} = 2$$

Equation of tangent plane

$$8(x - 2) + 2(y - 1) + 2(z + 3) = 0$$

$$4x + y + z - 6 = 0$$

Equation of normal is

$$\frac{x-2}{8} = \frac{y-1}{2} = \frac{z+3}{2}$$

2. Find the equation of the tangent plane and normal line to the surface $x^2 + 4y^2 = z^2$ at the point (3, 2, 5).

Solution

We have, $f(x, y, z) = x^2 + 4y^2 - z^2 = 0$

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 8y, \quad \frac{\partial f}{\partial z} = -2z$$

At point (3, 2, 5)

$$\frac{\partial f}{\partial x} = 6, \quad \frac{\partial f}{\partial y} = 16, \quad \frac{\partial f}{\partial z} = -10$$

Equation of tangent plane

$$6(x - 3) + 16(y - 2) - 10(z - 5) = 0$$

$$6x + 16y - 10z = 0$$

Equation of normal is

$$\frac{x-3}{6} = \frac{y-2}{16} = \frac{z-5}{-10}$$

3. Find the equation of the tangent plane and normal line to the surface $x^2 + y^2 + z^3 + xy - 2x + yz = 7$ at the point (0, 2, 1).
 4. Find the equation of the tangent plane and normal line to the surface $x^2 + y^2 + z^3 = 7$ at the point (1, 1, 1).

Discussion Questions

1. What is a Tangent's Plane?
2. Are tangents and normals perpendicular to each other?
3. How many tangents can be drawn to a surface?
4. How many points at which the tangent plane touches the surface?